

Calculus I Lesson 12

1) $f(x) = x + 2$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ $f'(x)$ is called the derivative of $f(x)$

2) $f(x) = 3x + 2$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ $f'(x)$ is called the derivative of $f(x)$

3) $f(x) = -4x + 1$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ $f'(x)$ is called the derivative of $f(x)$

4) $f(x) = x^2 + 2$

Hint: $(x + \Delta x)^2 = (x + \Delta x)(x + \Delta x) = x^2 + x \cdot \Delta x + x \cdot \Delta x + (\Delta x)^2 = x^2 + 2x \cdot \Delta x + (\Delta x)^2$

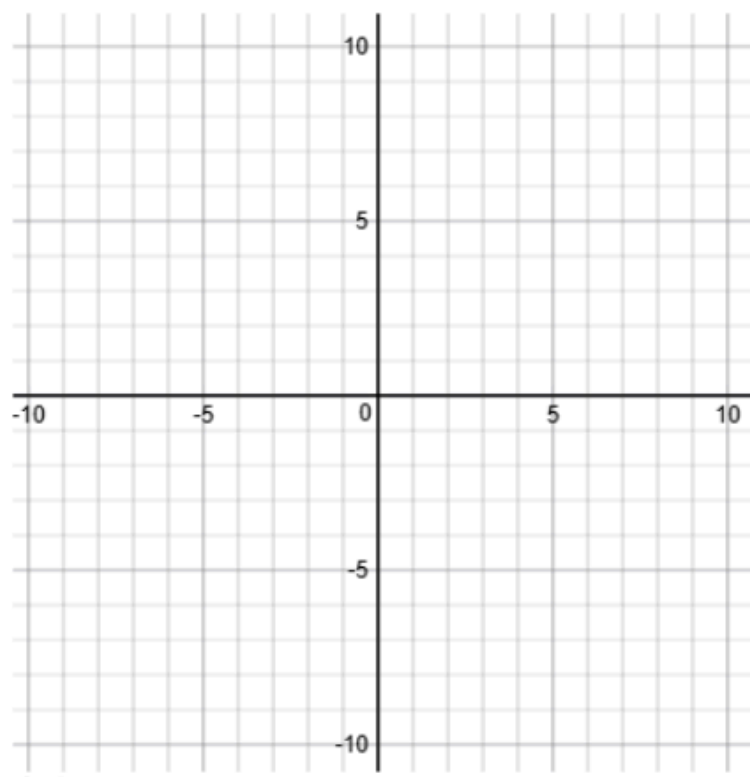
a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$f'(x)$ is called the derivative of $f(x)$

c) $f(1) = \underline{\quad ? \quad}$ and $f'(1) = \underline{\quad ? \quad}$

d) Graph $f(x) = x^2 + 2$ and the line $y = f'(1) \cdot (x - 1) + 3$



5) $f(x) = 2x^2 - 4$

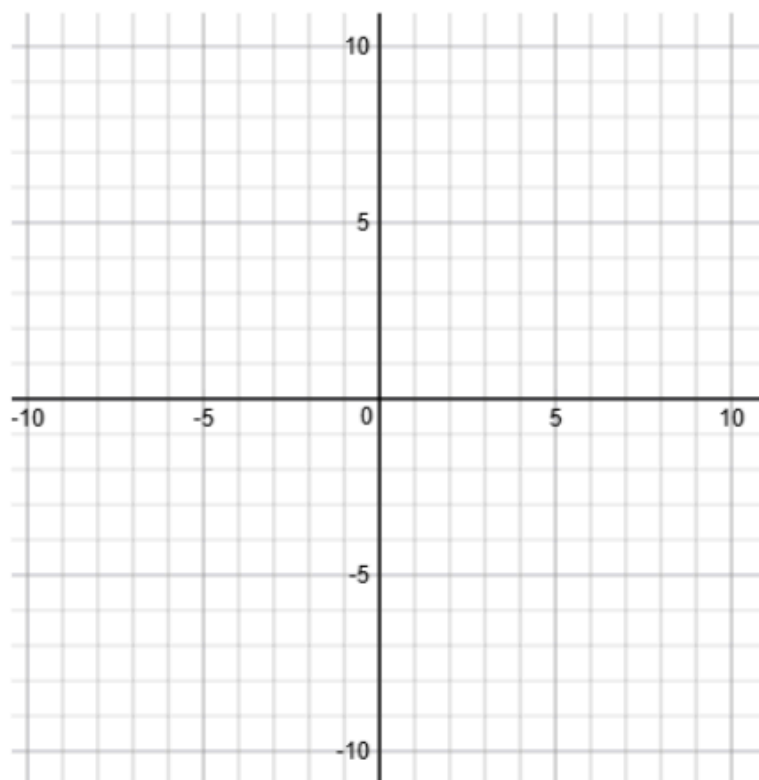
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b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$f'(x)$ is called the derivative of $f(x)$

c) $f(1) = \underline{\quad ? \quad}$ and $f'(1) = \underline{\quad ? \quad}$

d) Graph $f(x) = x^2 + 2$ and the line $y = f'(1) \cdot (x - 1) + 2$



6) $f(x) = x^3$

Hint: $(a + b)^3 = a^3 + 3a^2 \cdot b + 3a \cdot b^2 + b^3$

$$(x + \Delta x)^3 = x^3 + 3x^2 \cdot \Delta x + 3x \cdot (\Delta x)^2 + (\Delta x)^3$$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ $f'(x)$ is called the derivative of $f(x)$

c) $f(1) = \underline{\quad ? \quad}$ and $f'(1) = \underline{\quad ? \quad}$

d) Graph $f(x) = x^3$ and the line $y = f'(1) \cdot (x - 1) + 1$

7) $f(x) = 0.5x^3 + 1$

Hint: $(a + b)^3 = a^3 + 3a^2 \cdot b + 3a \cdot b^2 + b^3$

$$(x + \Delta x)^3 = x^3 + 3x^2 \cdot \Delta x + 3x \cdot (\Delta x)^2 + (\Delta x)^3$$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) Find $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ $f'(x)$ is called the derivative of $f(x)$

c) $f(2) = \underline{\quad ? \quad}$ and $f'(2) = \underline{\quad ? \quad}$

d) Graph $f(x) = 0.5x^3 + 1$ and the line $y = f'(2) \cdot (x - 2) + 5$

